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Letter

Ni-Catalyzed Synthesis of Thiocarboxylic Acid Derivatives

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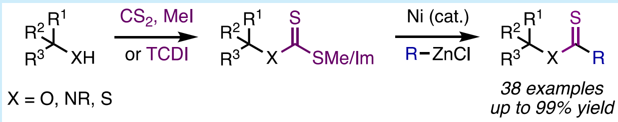
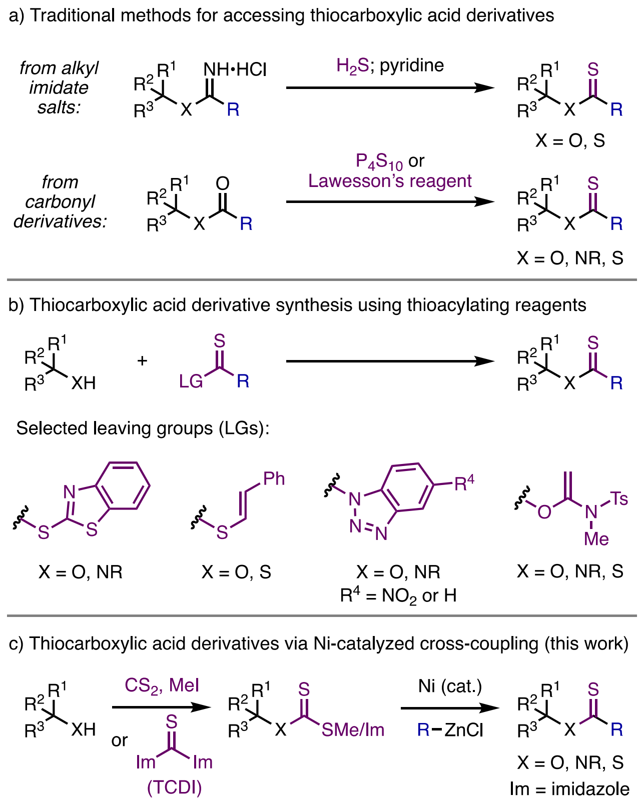
ABSTRACT: A Ni-catalyzed cross-coupling of readily accessible *O*-alkyl xanthate esters or thiocarbonyl imidazolides and organozinc reagents for the synthesis of thiocarboxylic acid derivatives has been developed. This method benefits from a fast reaction time, mild reaction conditions, and ease of starting material synthesis. The use of transition-metal catalysis to access a diverse range of thiocarbonyl-containing compounds provides a useful complementary approach when compared with previously established methodologies.



Thiocarboxylic acid derivatives such as thionoesters, thioamides, and dithioesters are an important class of compounds with a diverse range of applications.[1] For instance, the unique properties associated with the thiocarbonyl moiety enables selective reduction, dimerization, and aldol or Michael reactions, which have been leveraged in many total synthesis efforts.[2−10] The thiocarbonyl group can also serve as a convenient precursor for the CF_2 functional group,[11] which can impart desirable effects on pharmaceuticals[12] and agrochemicals.[13] Thiocarboxylic acid derivatives have also been applied toward the synthesis of a large array of sulfur-containing heterocycles[14−16] and are useful 1,2-[17] or 1,3-dipolarophiles[18−20] in cycloaddition reactions. From a biological perspective, thionoesters and dithioesters have recently shown promise as chemical agents that can slowly release hydrogen disulfide (H_2S), an important signaling molecule, under physiological conditions in investigations aimed toward the development of novel thiol-triggered H_2S donors in biological systems.[21,22] It is also known that the substitution of native ester and amide bond linkages in biologically active peptides with the corresponding thiocarbonyl analogue can modulate the conformation, proteolytic behavior, and activity of these peptides.[23−25]

Despite the use of thiocarboxylic acid derivatives in organic and biological chemistry, accessing these products can be challenging. Historically, thiocarboxylic acid derivatives have been accessed via the hydrolysis of an alkyl imidate salt, generated from the corresponding alcohol/thiol via the Pinner reaction, with hydrogen sulfide and pyridine.[26−28] More frequently, these compounds are accessed by treating the corresponding carbonyl-containing compound with a thionating agent such as phosphorus decasulfide (P_4S_{10})[29] or Lawesson's reagent[30] (Scheme 1a). These methods suffer from a range of drawbacks that include having to handle gaseous or malodorous reagents. They can also suffer from selectivity issues during thionoester or dithioester synthesis.[27−30] Thionation procedures, in particular, can involve long reaction times, high temperatures, and a challenging

Scheme 1. Synthesis of Thiocarboxylic Acid Derivatives



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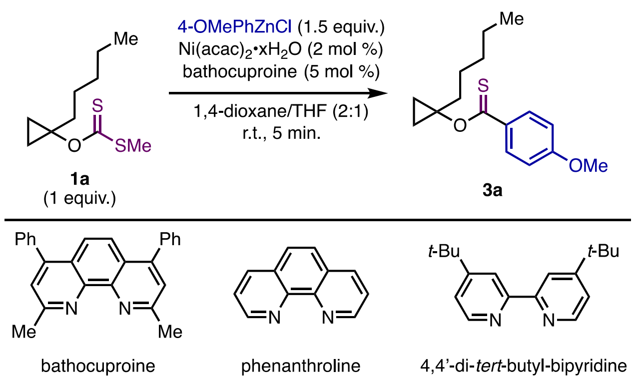
purification of the reaction product from reaction by-products.[29,30] The use of thioacyl chlorides or thioketene reagents as an alternative to these traditional routes is less commonly pursued due to the instability of these reagents, which makes their synthesis and storage difficult.[31,32] To address these pitfalls, more recent approaches for the synthesis of thiocarboxylic acid derivatives have been developed using a range of independently synthesized thioacylating agents (Scheme 1b, various leaving groups (LGs)).[33−43] These methods provide an alternative strategy to access thionoester, thioamide, or dithioester products under comparatively milder reaction conditions. However, these strategies often require multiple nontrivial additional synthetic steps[33−37] or the use of traditional methods (e.g., use of a thionating agent) to install the requisite thiocarbonyl moiety.[38−42]

There remains a need for the development of an efficient and general synthetic method to access thiocarboxylic acid derivatives from readily accessible thiocarbonyl-containing feedstocks under mild conditions. Herein, we report a Ni-catalyzed Negishi cross-coupling of *O*-alkyl xanthate esters and thiocarbonyl imidazolides for the generation of a diverse array of thiocarboxylic acid derivatives (Scheme 1c). This method notably employs an underexplored retrosynthetic disconnection relative to conventional methodologies (Scheme 1)[44,45] and benefits from a fast reaction time, low catalyst loadings, and simple starting material synthesis.

Inspired by findings in our recent work focused on the Ni-catalyzed C(sp^3)−O arylation of tertiary cyclopropanols using a redox-active thiocarbonyl-containing leaving group,[46] we selected *O*-alkyl xanthate ester **1a** and 4-OMePhZnCl (**2a**) as the organozinc reagent as a starting point for the reaction optimization. The optimal reaction conditions discovered for this transformation employed Ni(acac)$_2$·xH$_2$O (2 mol %) and bathocuproine (5 mol %) in 1,4-dioxane/tetrahydrofuran (THF) (2:1) at room temperature for 5 min. Under these conditions, the desired product **3a** was obtained in 93% isolated yield (Table 1, entry 1). It was found that specific Ni(II) precatalysts or bidentate ligands could be substituted to obtain similar desired product yields (entries 3 and 4), whereas other substitutions performed less efficiently in comparison (entries 2 and 5). The replacement of 1,4-dioxane with THF or decreasing the amount of arylzinc reagent to 1.1 equiv resulted in a moderate decrease in the yield of the desired product (entries 6 and 7). Variation of the ratio of 1,4-dioxane/THF to 3:1 also slightly decreased the product yield (entry 8), whereas a 1:1 ratio was comparable to the optimal reaction conditions (entry 9). This reaction could notably be performed with decreased Ni(II) precatalyst (1 mol %) and bathocuproine (2 mol %) loadings without a significant effect on the reaction performance (entry 10). Finally, we found that both Ni and ligand are required for this reaction (entries 11 and 12).

The scope of thionoesters that could be prepared using this transformation was evaluated using the optimal reaction conditions with the goal of exploring a diverse range of alcohol starting materials (Scheme 2). This protocol could be extended to another one-substituted tertiary cyclopropanol (**3b**) and unactivated secondary (**3c**−**3e**) and primary (**3f**−**3i**) alcohol derivatives. This reaction is also amenable to secondary (**3j**−**3l**) and primary (**3m**, **3n**) benzylic alcohol starting materials. An α-hydroxy ester (**3o**) xanthate was a competent substrate as well. This reaction notably tolerated a pyridine (**3g**), alkene (**3h**), terminal alkyne (**3i**), and aryl bromide (**3j**)

Table 1. Reaction Optimization[a]



entry	variation	yield[b]
1	none	99 (93)[c]
2	Ni(OAc)$_2$·4H$_2$O instead of Ni(acac)$_2$·xH$_2$O	58
3	NiCl$_2$·6H$_2$O instead of Ni(acac)$_2$·xH$_2$O	96
4	phenanthroline instead of bathocuproine	95
5	4,4′-di-*tert*-butyl-bipyridine instead of bathocuproine	72
6	THF instead of 1,4-dioxane	61
7	1.1 equiv of 4-OMePhZnCl	79
8	1,4-dioxane/THF (3:1)	85
9	1,4-dioxane/THF (1:1)	97
10	1 mol % Ni(acac)$_2$·xH$_2$O, 2 mol % bathocuproine	94
11	no Ni	0
12	no bathocuproine	4

[a]Reactions performed on a 0.10 mmol scale. See the SI for full details. [b]Gas chromatography−mass spectrometry (GC-MS) yield using *n*-dodecane as an internal standard. [c]Isolated yield.

and chloride (**3n**) functional groups. A thionoester containing a free secondary amide was obtained in moderate yield by increasing the arylzinc equivalents to 3.0 from 1.5 (**3e**). Finally, we found that the reaction was readily scaled to obtain the tertiary cyclopropanol thionoester **3a** in near-quantitative yield on a 1.0 mmol scale.

Next, we evaluated the reaction scope with respect to the arylzinc reagent (Scheme 3). We found that other electron-rich (**3p**, **3t**) and electron-deficient (**3s**) arylzinc reagents could be used to obtain thionoester products in high yields. In addition, product **3q**, resulting from the cross-coupling of a naphthyl zinc reagent, and products **3r** and **3u**, resulting from the cross-coupling of heteroaryl zinc reagents, were obtained in moderate to high yields. Ortho-substituted nucleophiles were not well tolerated under these reaction conditions (see the SI for details).

On the basis of the successful formation of a broad range of aryl thionoester products from *O*-alkyl xanthate esters, we wondered if this strategy could be applied toward the synthesis of other thiocarboxylic acid derivatives. After brief optimization (see the SI for details), we found that secondary amines, following conversion to the corresponding thiocarbonyl imidazolide **4**, could be used as starting materials for the generation of aryl thioamide products. By switching the ligand to phenanthroline (5 mol %) and extending the reaction time to 1 h, a diverse range of tertiary aryl thioamide products were obtained (Scheme 4).[47] Thiocarbonyl imidazolides derived from acyclic and cyclic amines were compatible with this protocol, affording thioamide products **5a**−**c** and **5d**−**j**, respectively, in high yields. Indoline (**5h**) and tetrahydroquinoline (**5i**) thioamides were obtained in high yields. We found

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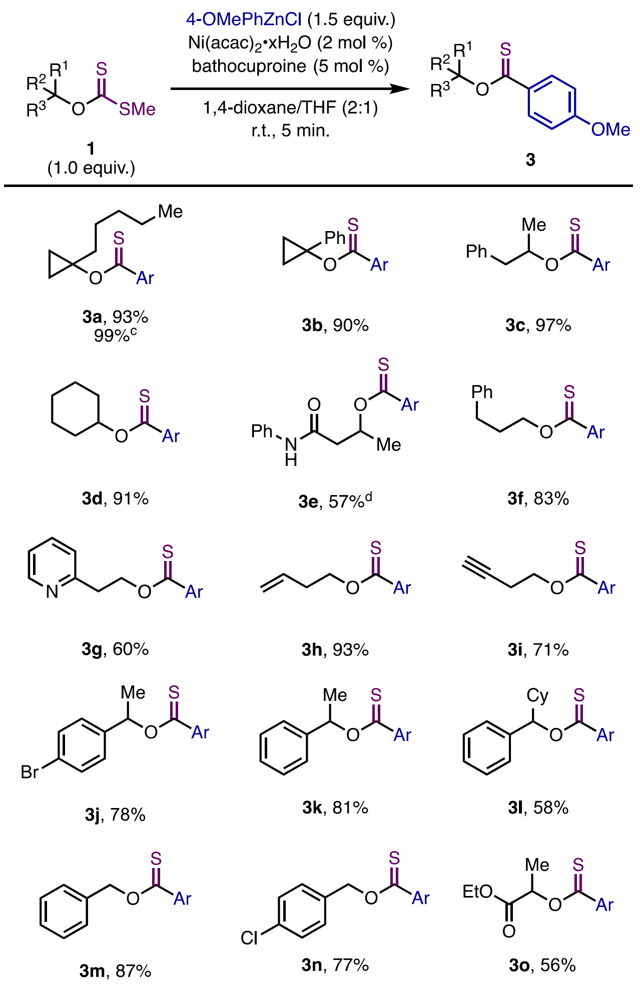
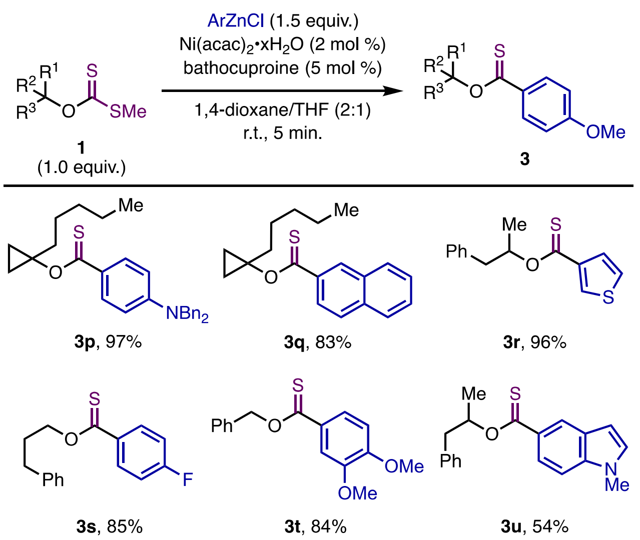
Scheme 2. Thionoester Scope[a,b]



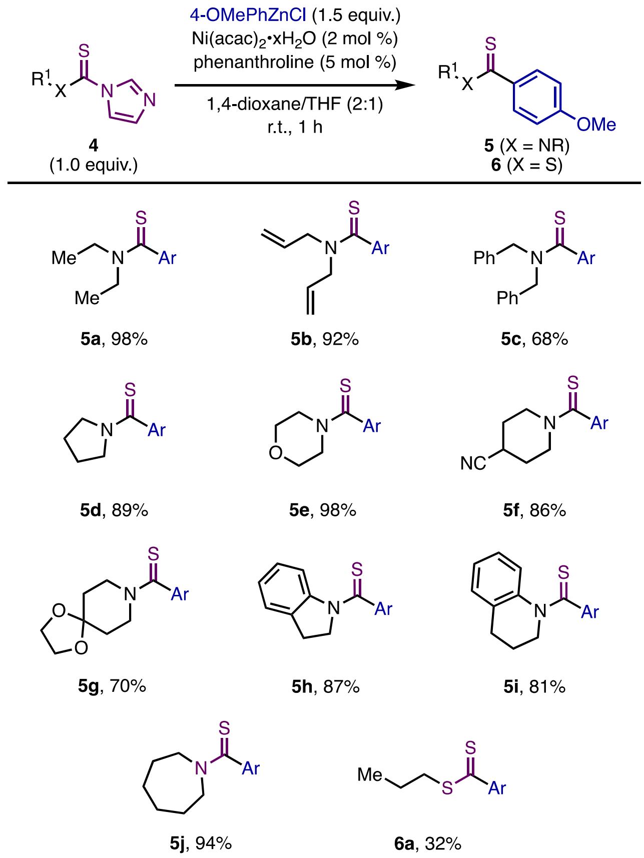
[a]Reactions performed on a 0.20 mmol scale. See the SI for full details [b]Isolated yield [c]Reaction performed on a 1.0 mmol scale [d]3.0 equiv of aryl zinc.

Scheme 3. Arylzinc Scope[a,b]



[a]Reactions performed on a 0.20 mmol scale. See the SI for full details. [b]Isolated yield.

Scheme 4. Thioamide Scope[a,b]



[a]Reactions performed on a 0.20 mmol scale. See the SI for full details. [b]Isolated yield.

that allyl (**5b**), nitrile (**5f**), and ketal (**5g**) functional groups were tolerated. To examine if this protocol could be similarly extended toward the generation of analogous dithioester products, we applied the conditions used for thioamide formation to a thiocarbonyl imidazolide derived from 1-propanethiol to deliver the desired product **6a** in a modest 32% yield. This proof-of-concept example using unoptimized reaction conditions demonstrates that this strategy can also be extended toward dithioester formation.

To further demonstrate the breath of this approach toward the synthesis of thionoesters and thioamides, we briefly explored the use of an alkylzinc reagent as a coupling partner (Scheme 5). Minor modifications to the standard reaction conditions, including using $Ni(OAc)_2{\cdot}4H_2O$ as the precatalyst and terpyridine as the ligand, led to the identification of the optimal conditions for this transformation (see the SI for details). Ni-catalyzed cross-coupling of thiocarbonyl imidazolides **4** and *n*-pentyl alkylzinc led to thionoester and thioamide products in moderate to high yields. Substrates derived from primary (**3v**) and secondary (**3w, 3x**) alcohols and acyclic (**5k**) and cyclic (**5l, 5m**) secondary amines were amenable to these reaction conditions.

Since both *O*-alkyl xanthate ester and thiocarbonyl imidazole starting materials are easily prepared using the combination of carbon disulfide and methyl iodide, or 1,1′-thiocarbonyldiimidazole (TCDI), respectively, one-pot and telescoped protocols were investigated.[48] Both the one-pot and telescoped reactions to form thionoester products from alcohols gave modest yields of desired product (see the SI for details). However, analogous

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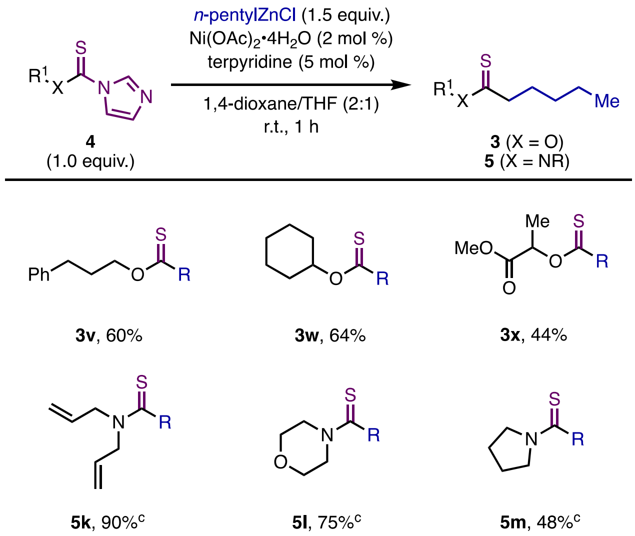
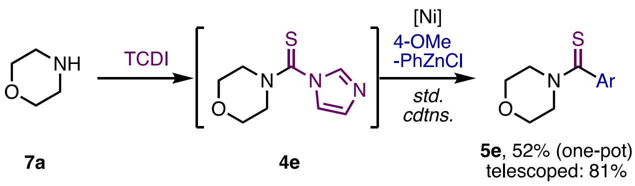
Scheme 5. *n*-Pentylzinc Chloride as a Coupling Partner[a,b]



[a]Reactions performed on a 0.20 mmol scale. See the SI for full details. [b]Isolated yield. [c]Performed using 2 mol % $Ni(acac)_2{\cdot}xH_2O$ and 5 mol % phenanthroline.

experiments to form thioamide **5e** using TCDI and morpholine (**7a**) were more efficient (Scheme 6). Following the *in situ*

Scheme 6. One-Pot and Telescoped Reactions to Prepare Thioamide 5e[a,b]



[a]Reactions performed on 0.30 mmol scale. See the SI for full details [b]Isolated yield.

generation of thiocarbonyl imidazolide **4e** in a one-pot protocol, the standard reaction conditions gave aryl thioamide product **5e** in a moderate 52% yield (cf. 98% yield for the standard protocol starting from **4e** on a 0.2 mmol scale). The telescoped protocol for aryl thioamide synthesis was more efficient, giving the product in 81% yield. For those focused on rapidly generating a breadth of thiocarboxylic derivatives, the ability to conduct either a one-pot or telescoped procedure from free alcohol or amine starting materials and commercially available thiocarbonyl sources should be of interest despite the observed diminished reaction efficiency.

In summary, we have developed a Ni-catalyzed cross-coupling of organozinc reagents and thiocarbonyl containing starting materials to access an array of thionoester, thioamide, and dithioester derivatives. The ease of starting material preparation using readily accessible carbon disulfide or 1,1′-thiocarbonyldiimidazole as thiocarbonyl sources combined with mild reaction conditions, low catalyst loadings, and fast reaction times renders this method an attractive alternative for the synthesis of thiocarboxylic acid derivatives. The use of thiocarbonyl-containing materials in other transition-metal-catalyzed cross-coupling reactions is currently being investigated in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c04074.

Reaction optimization tables, mechanistic studies, synthetic procedures, characterization data, and NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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